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Linealidad de la integral

Teorema 1 (Linealidad de la integral). *Sea* (X, Σ, μ) *un espacio de medida,* $E \in \Sigma$ *y* $f, g\colon X \longrightarrow \mathbb{R}$ *funciones integrables en* E *(es decir,* $f, g \in \mathcal{L}^1(\mu, E)$*)*

$$\implies \forall a, b \in \mathbb{R} : \int_E (af + bg)\,\mathrm{d}\mu = a \int_E f\,\mathrm{d}\mu + b \int_E g\,\mathrm{d}\mu$$

***Demostración*:** Veamos primero que $\forall a \in \mathbb{R} : \int af\,\mathrm{d}\mu = a \int f\,\mathrm{d}\mu$.

$$\text{por definición } \int af\,\mathrm{d}\mu = \int (af)^+\,\mathrm{d}\mu - \int (af)^-\,\mathrm{d}\mu$$

$$\text{Observamos que } \begin{cases} a \geqslant 0 \implies (af)^+ = af^+ \wedge (af)^- = af^- \\ a < 0 \implies (af)^+ = -af^- \wedge (af)^- = -af^+ \end{cases}$$

$$\implies \int (af)^+\,\mathrm{d}\mu - \int (af)^-\,\mathrm{d}\mu = \left.\begin{cases} a\int f^+\,\mathrm{d}\mu - a\int f^-\,\mathrm{d}\mu & \text{si } a \geqslant 0 \\ -a\int f^-\,\mathrm{d}\mu + a\int f^+\,\mathrm{d}\mu & \text{si } a < 0 \end{cases}\right\} = a \int f\,\mathrm{d}\mu$$

$$\implies \int (af + bg)\,\mathrm{d}\mu \overset{??}{=} \int af\,\mathrm{d}\mu + \int bg\,\mathrm{d}\mu = a \int f\,\mathrm{d}\mu + b \int g\,\mathrm{d}\mu$$

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Referenciado en

- `lem-esp-lp-vectorial`
- `prop-esperanza-fn`
- `obs-linealidad-esperanza`
- `teo-esp-l2-hilbert`
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